



AIRLINE LOYALTY PROGRAM

Data Engineering Project

DOCUMENTATION

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Table of Contents

Contents

Table of Contents	2
1. Data Overview	3
2. Logical & Physical Model	4
2.1 Conceptual Model	5
2.2 Logical Model.....	6
3. SSIS Overview.....	7
4. Power BI Overview.....	Error! Bookmark not defined.

1. Data Overview

This dataset simulates a **real-world airline loyalty program**, containing detailed information about member profiles, flight activity, and earned loyalty points across time. **It includes two core data files:**

1) Customers Flights Activity:

Monthly flight performance and loyalty point accumulation per customer, Each row represents a customer's flight activity for a single month •

Key Columns:

- a) **Loyalty Number:** Customer's unique loyalty number
- b) **Year:** Year of the period
- c) **Month:** Month of the period
- d) **Total Flights:** Sum of Flights Booked and Flights with Companions
- e) **Distance:** Flight distance traveled in the period (km)
- f) **Points Accumulated:** Loyalty points accumulated in the period
- g) **Points Redeemed:** Loyalty points redeemed in the period
- h) **Dollar Cost Points Redeemed:** Dollar equivalent for points redeemed in the period in CDN

2) Customers Loyalty History:

Each row describes a **snapshot of a customer's loyalty status** for a specific reporting period. • **Key Columns:**

- a) **Loyalty Number:** Customer's unique loyalty number
- b) **Country:** Country of residence
- c) **Province:** Province of residence
- d) **City:** City of residence
- e) **Postal Code:** Postal code of residence
- f) **Gender:** Gender
- g) **Education:** Highest education level (High school or lower > College > Bachelor > Master > Doctor)
- h) **Salary:** Annual income
- i) **Marital Status:** Marital status (Single Married Divorced)
- j) **Loyalty Card:** Loyalty card status (Star > Nova > Aurora)
- k) **CLV:** Customer lifetime value - total invoice value for all flights ever booked by member
- l) **Enrollment Type:** Enrollment type (Standard / 2018 Promotion)
- m) **Enrollment Year:** Year Member enrolled in membership program
- n) **Enrollment Month:** Month Member enrolled in membership program
- o) **Cancellation Year:** Year Member cancelled their membership
- p) **Cancellation Month:** Month Member cancelled their membership.

2. Logical & Physical Model

1) Schema Type: Star Schema

The model follows a **Star schema** structure with two central **fact tables** and several surrounding **dimension tables** that provide context. The schema is optimized for analytical querying and enables efficient slicing and dicing of the data across various perspectives.

2) Fact Tables:

- a. **FlightsActFact**: Captures monthly flight activity for each customer, including metrics like:
 - i. **TotalFlights**
 - ii. **Distance**
 - iii. **PointsAccumulated** iv. **PointsRedeemed**
 - v. **DollarCostPerPR**
- b. **LoyaltyHistFact**: Tracks periodic snapshots of customer loyalty performance:
 - i. **Salary** ii. **CLV (Customer Lifetime Value)**

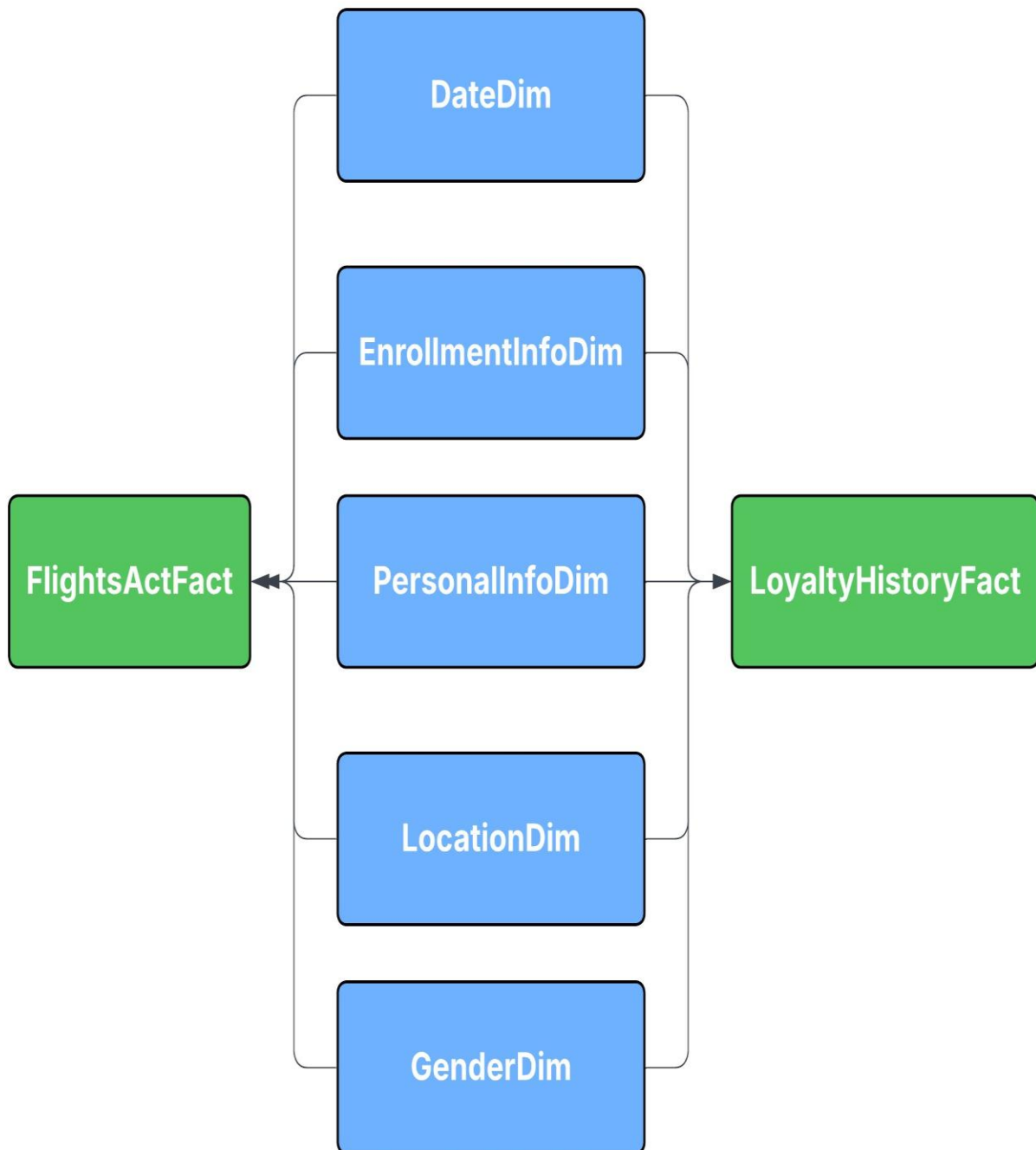
3) Dimensions Tables

- a. **DateDim** – Time-based dimension supporting analysis by Year, Month, Quarter, and Season.
- b. **GenderDim** – Classifies customers by gender.
- c. **LocationDim** – Provides geographical context like country, city, and postal code.
- d. **EnrollmentInfoDim** – Captures details about the loyalty card type (e.g., Star, Aurora, Nova) and enrollment type (e.g., Standard, Promotion 2018).
- e. **PersonallInfoDim** – Stores personal attributes such as education level and marital status.

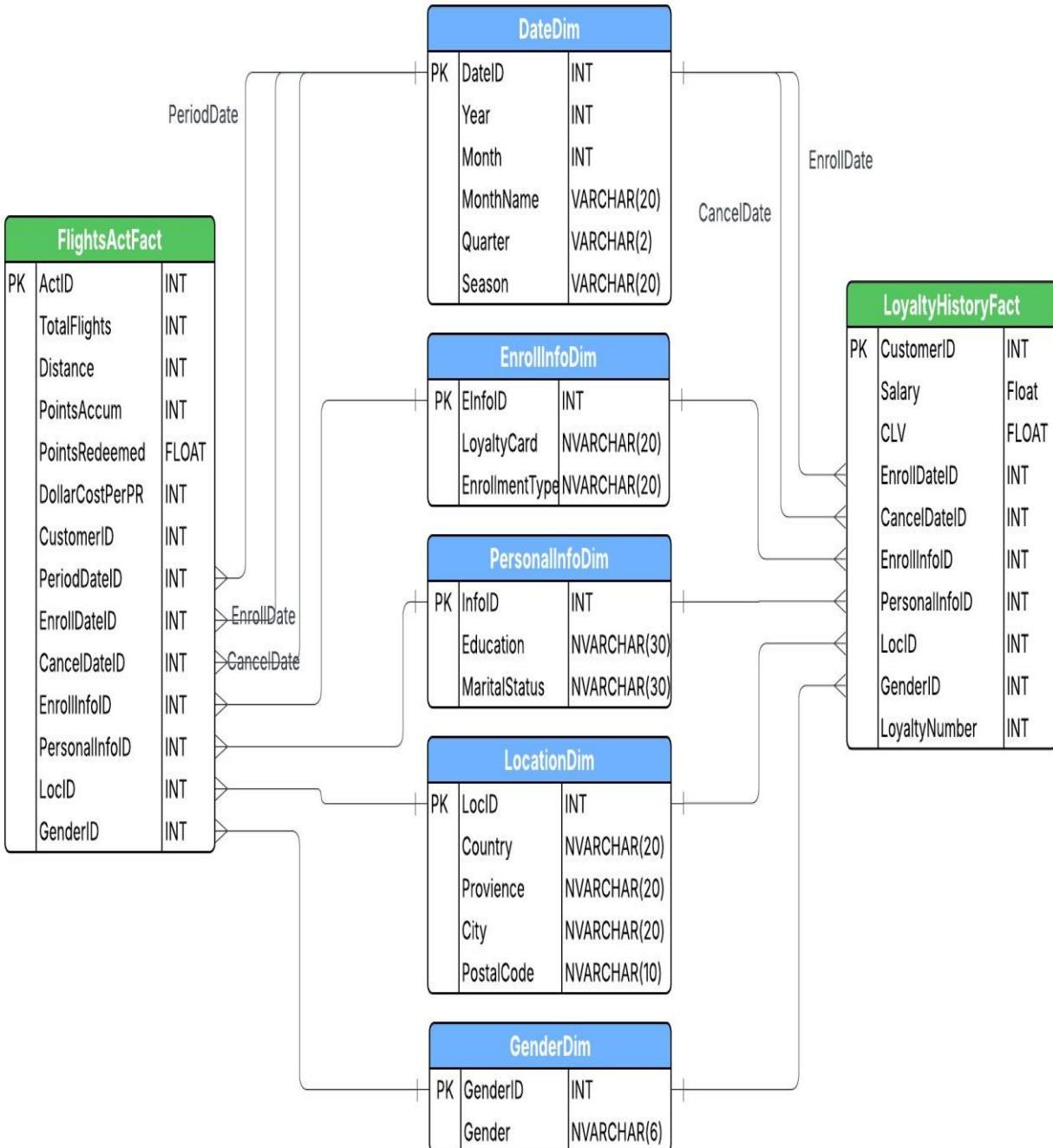
4) Key Relationships

- a. **FlightsActFact**: GenderDim, LocationDim, EnrollmentInfoDim, PersonallInfoDim, and Three roles of DateDim (for **Enrollment**, **Cancellation** and **Period Dates**).
- b. **LoyaltyHistFact**: GenderDim, LocationDim, EnrollmentInfoDim, PersonallInfoDim, and Three roles of DateDim (for **Enrollment**, **Cancellation** and **Period Dates**).

2.1 Conceptual Model



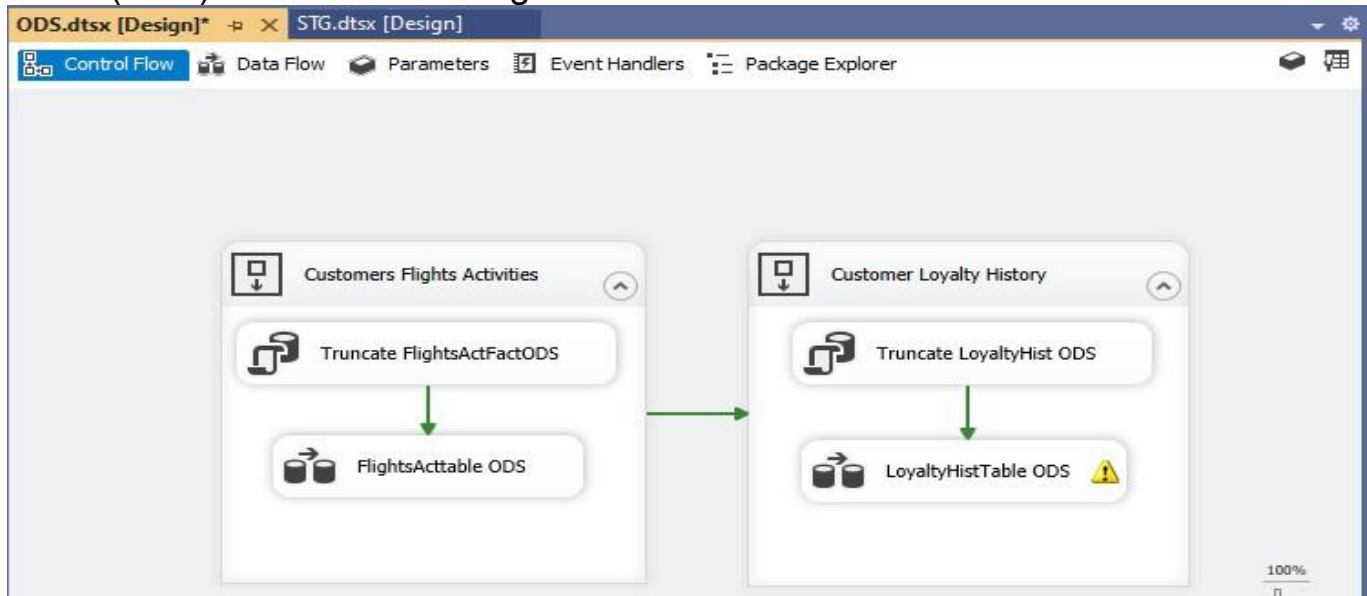
2.2 Logical Model



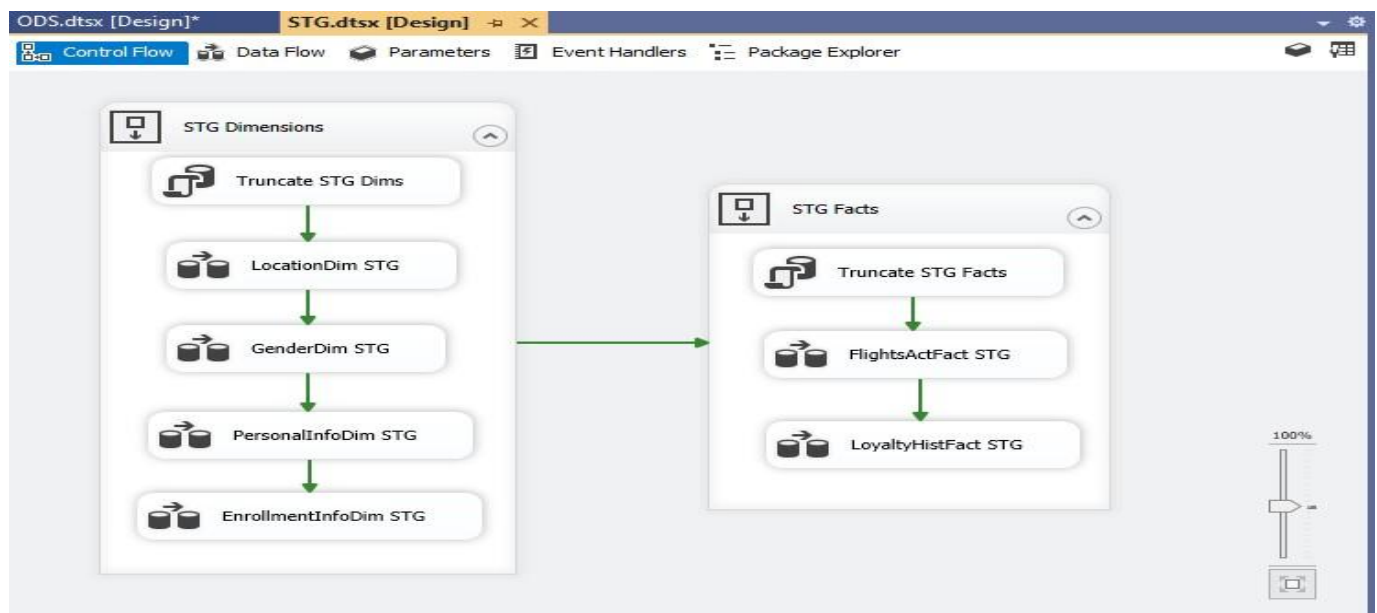
3. SSIS Overview

- In the **SSIS** Phase there are **3 packages** capturing the whole **ETL** Process for the project as follows:

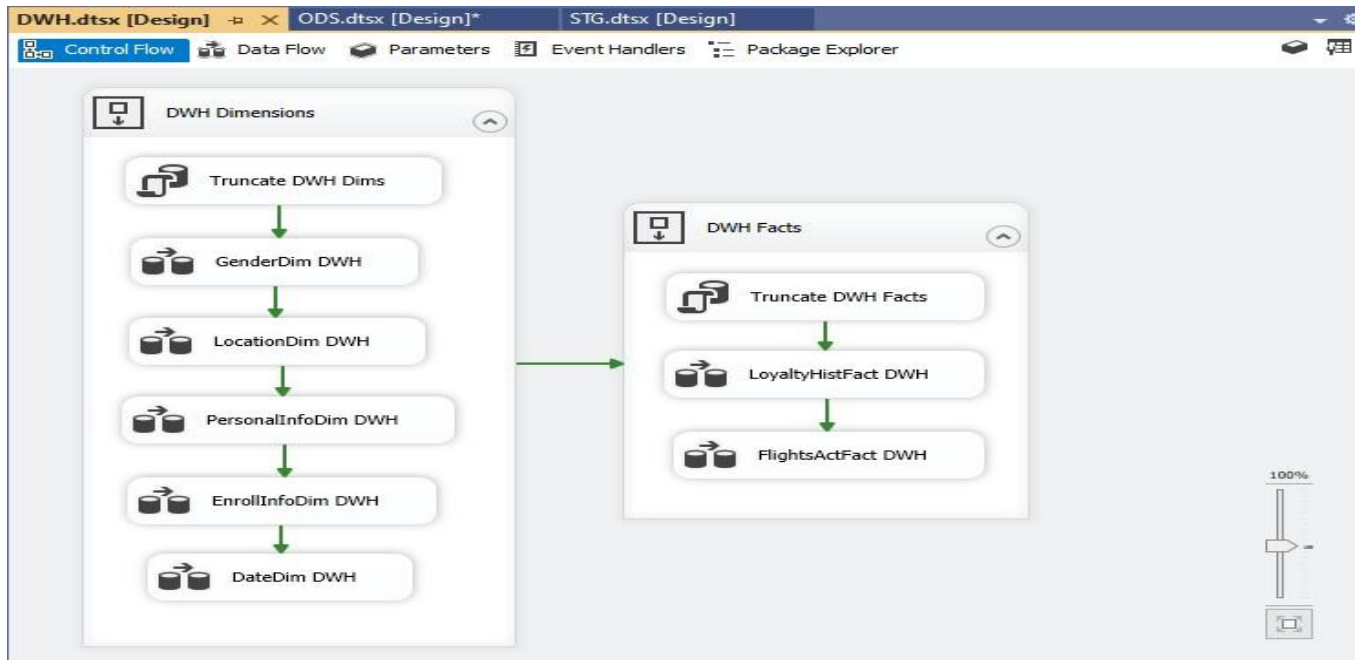
- Package 1: ODS** Package which handles the process of **Extracting** the data from the data sources which in this case are **Flat Files (CSV)** as shown in the figure below.



- Package 2: STG** Package Which involves Applying the Transformation and dimensions preparations for being ready to be uploaded to the DWH.



- C. Package 3: DWH** Where we Finalize the created facts and dimensions by getting lookups and filling the data warehouse with the data.



4. Power BI Overview

After Creating the semantic layer with its measures and relationships, it's time to Create the Power BI dashboard provides an interactive visualization layer for the data warehouse. It enables users to monitor key performance indicators, analyze customer loyalty trends, and explore flight and demographic data. Each report page is designed to deliver clear, actionable insights with user-friendly navigation and filtering options:

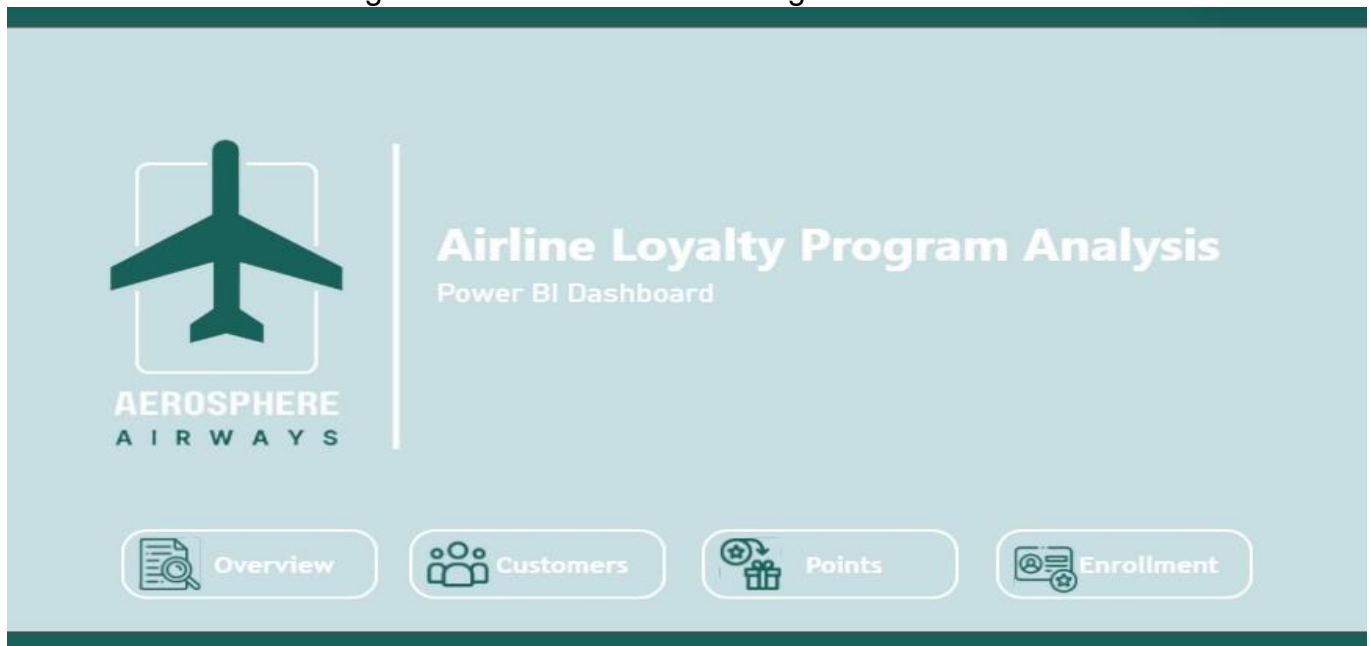
The created dashboard contains 5 pages as follows:

1. Home Page.
2. Data Overview Page.
3. Customers Analysis Page.
4. Points Analysis Page
5. Enrollment Analysis Page

These Pages are described and shows as follows:

1) Home Page:

- It is the page that shows the Project title and contains the Navigation Buttons to the other Pages within the Dashboard.



2) Overview Page:

- This Page contains a Historical Overview For the data like:
 - The History of Redeemed and Accumulated points over the years
 - The Enrollment and Cancellation for the customers to the program.
 - the program costs and the total revenue of the program.

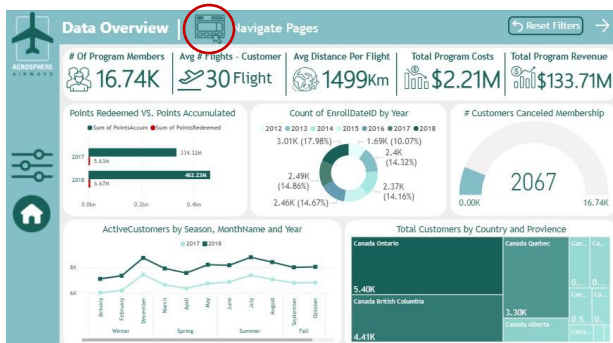


► The Slicers and Navigation for each page can be accessed through the setting and the navigation button exists in the page as follows.

► The Filters:



► The Page Navigation:

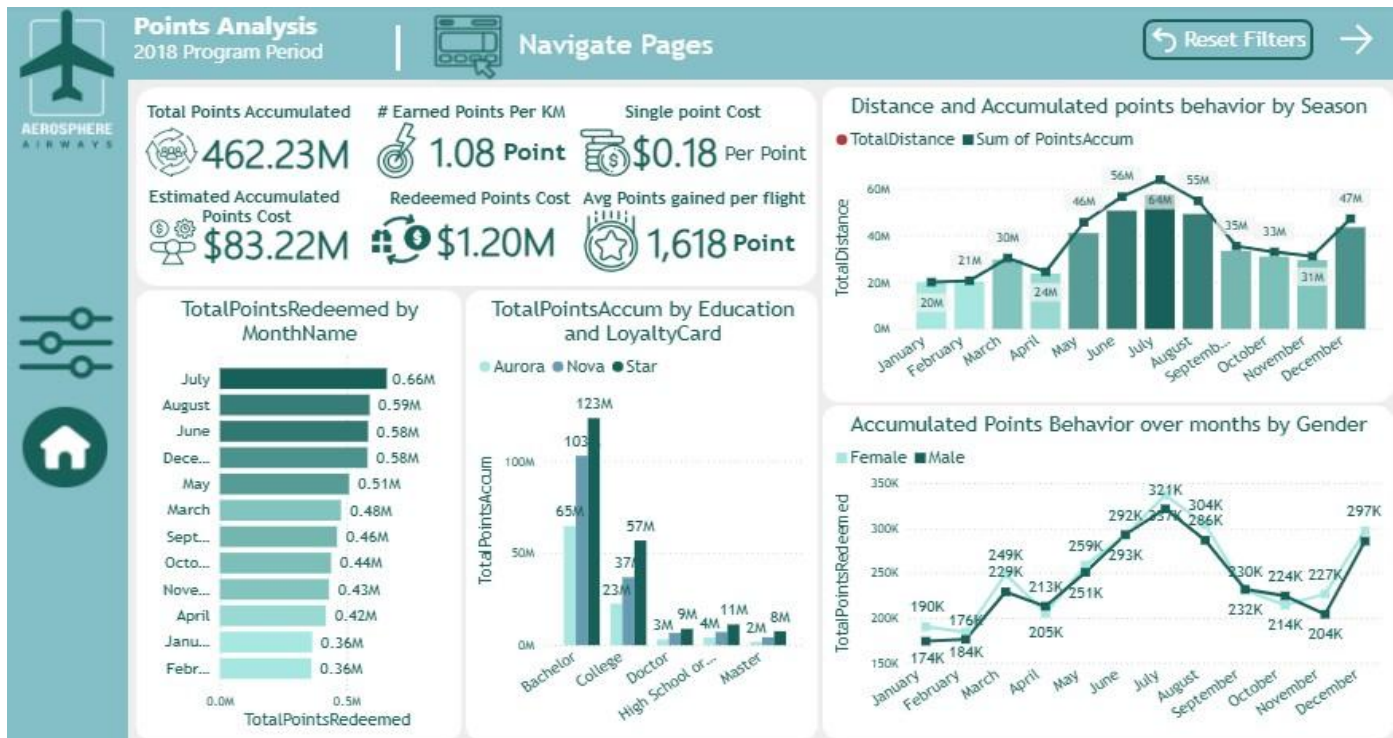


3) Customers Analysis Page

- The page contains visuals for the customers according to there demographics and there enrollment information.



4) Points Analysis Page:



5) Enrollment Analysis Page:

